BIOS 835: Linear Classification Methods

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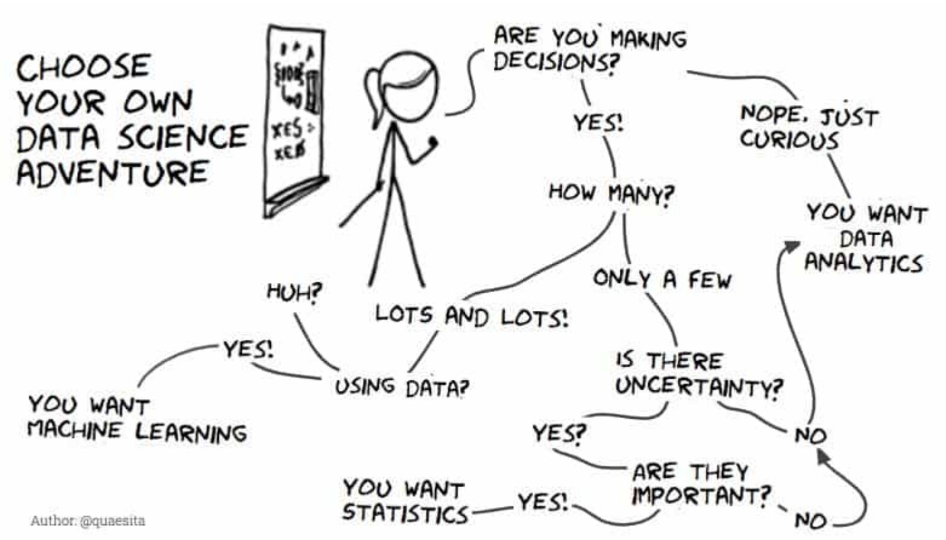
CHOOSE YOUR OWN DATA SCIENCE ADVENTURE
ARE YOU MAKING DECISIONS?
YES!
NOPE. JUST CURIOUS
HOW MANY?
YOU WANT DATA ANALYTICS
ONLY A FEW
HUH?
LOTS AND LOTS!
YES!
IS THERE UNCERTAINTY?
USING DATA?
YOU WANT MACHINE LEARNING
YES?
NO
ARE THEY IMPORTANT?
YOU WANT STATISTICS
YES!
NO
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Classification

Classification Introduction

- We'll now switch from continuous to binary (yes/no) types outcomes.
- For example, the Wisconsin Breast Cancer Dataset (WBCD) was created by Dr. William H. Wolberg at the University of Wisconsin Hospitals, Madison.
- These data are used to predict whether a breast cancer tumor is benign or malignant based on various cell features.

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- For example, the Wisconsin Breast Cancer Dataset (WBCD) was created by Dr. William H. Wolberg at the University of Wisconsin Hospitals, Madison.
- These data are used to predict whether a breast cancer tumor is benign or malignant based on various cell features.
- **Examples of cell features:**
  - Radius (mean of distances from the center to points on the perimeter),
  - Texture (standard deviation of gray-scale values),
  - Perimeter, Area, etc.

  Each of these features is calculated for three different metrics: mean, standard error, and the "worst" or largest value.

Classification Introduction

- Classification is the process of predicting the class label of a given input based on training data that contains input-output pairs.
- It involves building a model that learns from the training data and can predict the class of new, unseen data.
- Linear regression cannot be used for classification.
- Some Common Algorithms: K-Nearest Neighbors (KNN), logistic regression, decision trees, Naive Bayes, Neural Networks, Gradient Boosting Machines

Logistic Regression

- If we are classifying using logistic regression, we have

$$\pi_i = \Pr(Y_i = 1|\boldsymbol{x}_i) \quad = \quad \frac{\exp(\beta_0 + \boldsymbol{\beta}_1'\boldsymbol{x}_i)}{1 + \exp(\beta_0 + \boldsymbol{\beta}_1'\boldsymbol{x}_i)}$$

- The parameters are usually estimated using maximum likelihood methods, which give $\hat{\pi}_i$.

Model Selection in Logistic Regression

- The same questions about model selection come up.
- For example, which of the cell features should be used for our $\boldsymbol{x}$? Which summary measure should we use?
- The L_1 Lasso penalty discussed previously can be used for variable selection and shrinkage for the logistic regression model.

Model Selection Criteria

1. AIC/BIC
   - Akaike information criterion (AIC) and Bayesian information criterion (BIC) judge a model by the value of the negative log-likelihood ($-\log\{L(\beta)\} > 0$).
   - The formulas are

$$
\begin{aligned}
AIC &= -2\log\{L(\hat{\beta})\} + 2p \\
BIC &= -2\log\{L(\hat{\beta})\} + p\log(n)
\end{aligned}
$$

   - BIC penalizes more heavily for adding additional parameters.
   - We choose the model with the lowest AIC/BIC.
   - These can be used with forward/backward/step-wise selection procedures.

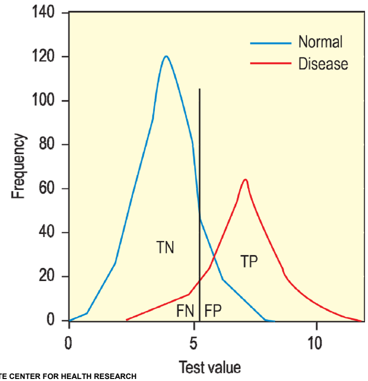
Measuring of predictive ability

- We will now take a bit of a tangent to discuss quantifying the predictive ability of a model.
- Some of these methods will take more background than others.
- In general, they are all trying to answer the following:

  Ten new subjects walk into the room (all were not in the data used to estimate $\hat{\beta}$), which model is going to give me the best estimate of the predictive probability (i.e., $\hat{\pi}_i$) for those ten subjects?

Sensitivity and specificity

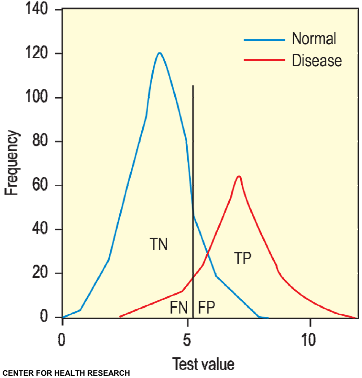
- Imagine a study evaluating a new test that screens people for a disease $\{0, 1\}$.
- The test outcome can be positive (predicting that the person has the disease) or negative (predicting that the person does not have the disease).
- The test results for each subject may or may not match the subject's actual status.



Sensitivity and specificity

In this setting, there are 4 things that could happen:

- **True positive (TP):** Sick people correctly diagnosed as sick
- **False positive (FP):** Healthy people incorrectly identified as sick
- **True negative (TN):** Healthy people correctly identified as healthy
- **False negative (FN):** Sick people incorrectly identified as healthy



2x2 Classification

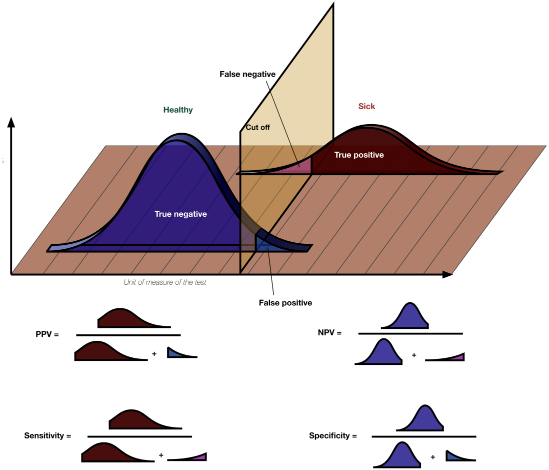
	Test Pos. (T+)	Test Neg. (T-)	Totals
Diseased (D+)	TP	FN	TP+FN
Not diseased (D-)	FP	TN	FP+TN
Totals	TP+FP	FN+TN	n

Where:

- TP+FN: **number of sick people.**
- FP+TN: **number of health people.**
- TP+FP: number that tested positive.
- TN+FN: number that tested negative.

We'll use FP, FN, FP and TN to quantify the predictive value model.

Sensitivity, specificity, PPV, NPV



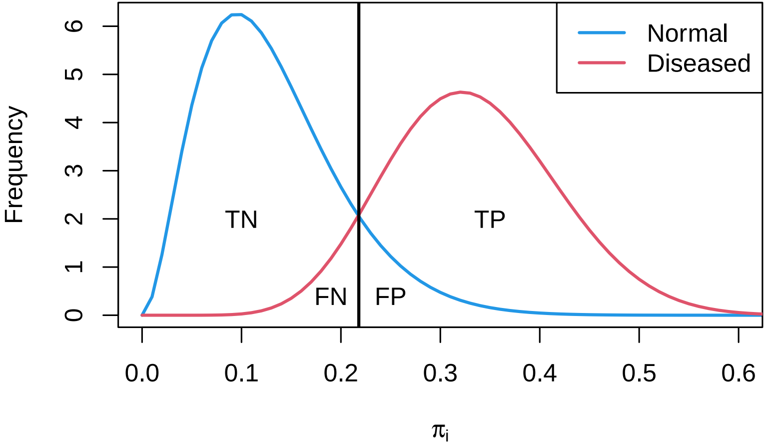
Estimating sensitivity and specificity from a logistic model

- To estimate sensitivity and specificity from a logistic model we we'll use the values of $\hat{\pi}_i$ as the results of the "test."
- Let's assume that $Y = 1$ if the person has the disease, so that a high value of $\hat{\pi}_i$ is indicative that the person will have the disease.
- Then we can pick a value, say π_0, such that the result of the test is

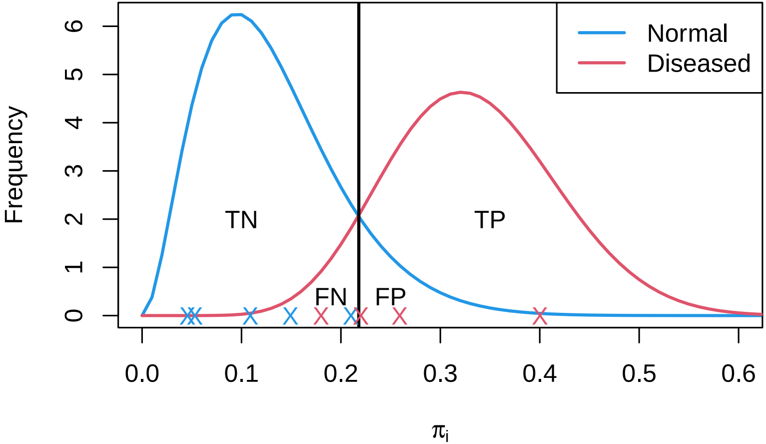
$$T_i = \begin{cases} +, & \text{if } \hat{\pi}_i > \pi_0 \\ -, & \text{if } \hat{\pi}_i \leq \pi_0 \end{cases}$$

- So all people with a predicted value of at least π_0 will have a test equal to "positive".

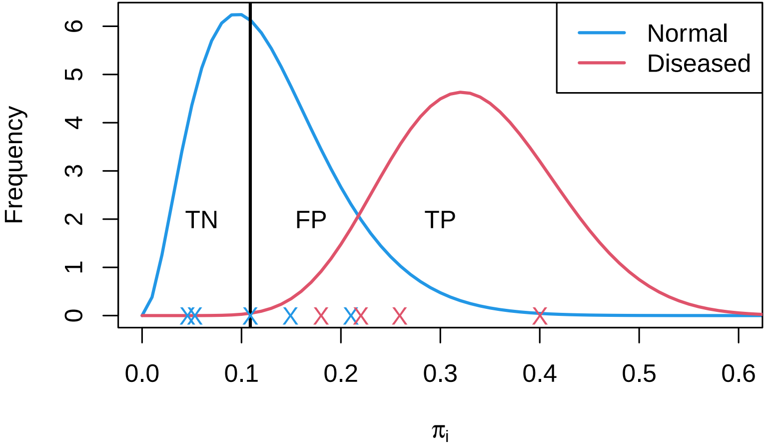
Classification of a test



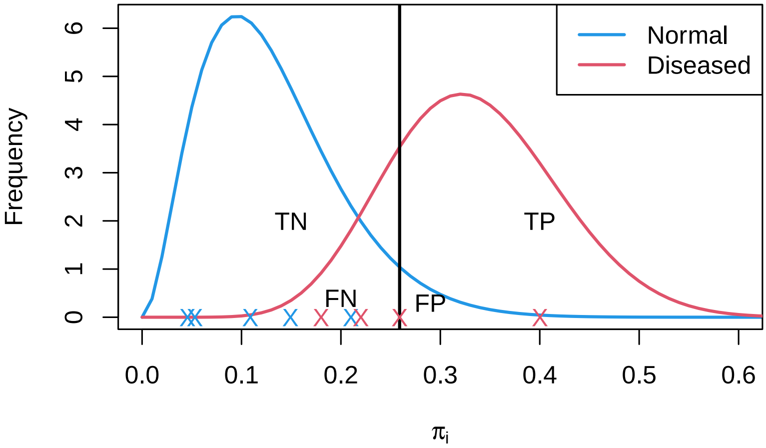
Classification of a test



Classification of a test



Classification of a test



Confusion Matrix and π_0

For each value of π_0

$$T_i = \begin{cases} +, & \text{if } \hat{\pi}_i > \pi_0 \\ -, & \text{if } \hat{\pi}_i \leq \pi_0 \end{cases}$$

we get a different confusion matrix:

	T+	T-	Totals
D+	TP(π_0)	FN(π_0)	# Diseased
D-	FP(π_0)	TN(π_0)	# Not Diseased
Totals	# Pos tests(π_0)	# Neg tests(π_0)	n

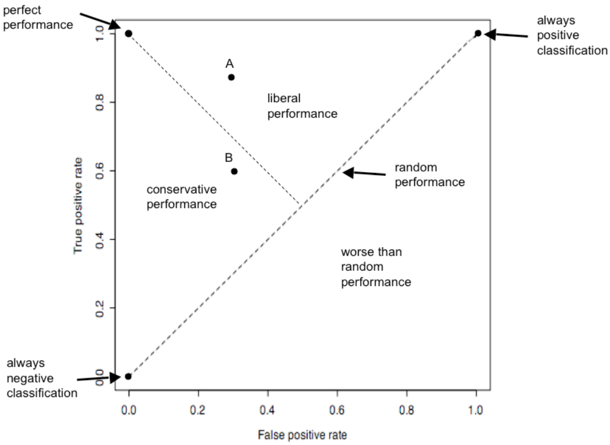
- The number of negative tests increases as π_0 increases.
- The number of positive tests decreases as π_0 increases.
- When $\pi_0 = 0$, the # Pos tests(0) $= n \rightarrow TN(0) = 0$
- When $\pi_0 = 1$, the # Neg tests(1) $= n \rightarrow TP(1) = 0$

Receiver Operating Characteristic (ROC) Curves

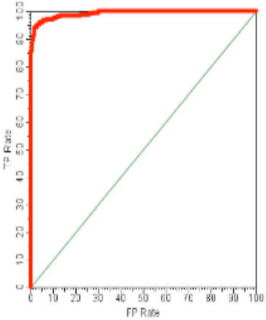
- Receiver Operating Characteristic (ROC) Curves are a way to measure the predictive ability of a model.
- ROC curves look at the $Sens(\pi_0)$ and $1 - Spec(\pi_0)$ for all values of π_0.
- When $\pi_0 = 0$ all tests are positive, $Sens(0) = 1$ and $1 - Spec(1) = 1$.
- When $\pi_0 = 1$ all tests are negative, $Sens(1) = 0$ and $1 - Spec(1) = 0$.

Motivation for ROC Curves

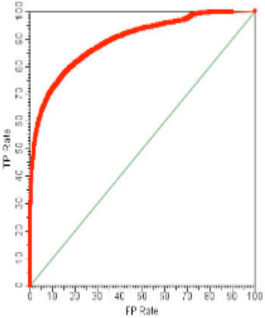
- For example, assume we have 100 diseased and 100 not diseased in our sample.
- Suppose that we list all 200 people from smallest to largest $\hat{\pi}_i$.
- Suppose that when I increase π_0 from 0.4 to 0.5 an additional 20 people have T^-.
- If the test is worthless I would expect 10 of those people to be diseased and 10 not diseased.
- In general, (If the test is worthless) as π_0 increases, 50% of the people are diseased and 50% are not diseased.


perfect performance
always positive classification
A
liberal performance
B
random performance
conservative performance
worse than random performance
always negative classification
True positive rate
0.0
0.2
0.4
0.6
0.8
1.0
False positive rate

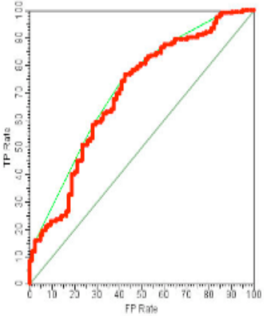
ROC curve examples



Almost perfect prediction



Good prediction



Poor prediction

Quantifying ROC curve performance

- Notice that the models that predict better have larger area under the ROC curve (AUC).
- AUC is the main way to quantify an ROC curve.
- The AUC is equal to the probability that the model will rank a randomly chosen diseased individual higher than a randomly chosen healthy individual.

$$\Pr(\hat{\pi}_i > \hat{\pi}_j | Y_i = 1 \text{ and } Y_j = 0)$$

AUC values

- We will interpret the AUC in terms of the discriminative ability of the model.
- That is, how well does the model discriminate between diseased and non-diseased individuals?

$\widehat{AUC}$	Interpretation of discriminative ability
< 0.5	worse than expected by chance
$= 0.5$	no discrimination
$0.7 - 0.8$	acceptable discrimination
$0.8 - 0.9$	excellent discrimination
> 0.9	outstanding discrimination

Cross-Validation for Logistic Regression Models

- Suppose we have several candidate logistic regression models (different sets of predictors, different functional forms).
- To compare them fairly, we need to evaluate performance on data not used to fit the model.
- **K-fold Cross-Validation:**
  - Split the data into K folds.
  - Each fold is the validation set once, with the others used for training.
  - Average the results across folds.
- For each model, we can compute performance metrics in each fold:
  - AUC
  - Sensitivity, Specificity
  - Positive Predictive Value (PPV), Negative Predictive Value (NPV)

Using Metrics to Select the Best Model

- After cross-validation, each candidate model has averaged performance metrics.
- **Choose the model** based on:
  - Highest AUC (if overall discrimination is the priority).
  - Best sensitivity/specificity balance (if identifying positives/negatives is critical).
  - Highest PPV/NPV (if predictive value matters most in the public health context).
- Key point: the "best" model depends on which metric aligns with the study goal.
- Example:
  - For screening, prioritize sensitivity/NPV.
  - For confirming disease, prioritize specificity/PPV.
- Cross-validation ensures these metrics reflect how the model will perform on new data.

MLE Logistic Regression

- In logistic regression, $\boldsymbol{\beta}$ can be estimated by maximizing the log-likelihood

$$\log\{L(\boldsymbol{\beta})\} = \sum_{i=1}^{N} \log\left\{\pi_i^{y_i}(1-\pi_i)^{y_i}\right\}$$

- This can be written in terms of $\boldsymbol{\beta} = \{\beta_{01}, \boldsymbol{\beta}_1\}$ as

$$\log\{L(\boldsymbol{\beta})\} = \sum_{i=1}^{N} \left\{y_i\boldsymbol{\beta}'\boldsymbol{x}_i - \log(1 + e^{\boldsymbol{\beta}'\boldsymbol{x}_i})\right\}$$

  where $\boldsymbol{x}$ now contains an intercept term.
- $\boldsymbol{\beta}$ is usually estimated using an algorithm called *iteratively reweighted least squares* or IRLS.

L_1 penalized logistic regression

- The L_1 Lasso penalty discussed previously can be used for variable selection and shrinkage.
- For the 2-class case, we look to find

$$\max_{\boldsymbol{\beta}} \left[\sum_{i=1}^{N} \left\{ y_i \boldsymbol{\beta}' \boldsymbol{x}_i - \log(1 + e^{\boldsymbol{\beta}' \boldsymbol{x}_i}) \right\} - \lambda \sum_{j=1}^{p} |\beta_j| \right]$$

- This is a difficult concave optimization, but the IRLS solution can be applied.
- This leads to a repeated application of a weighted lasso algorithm.

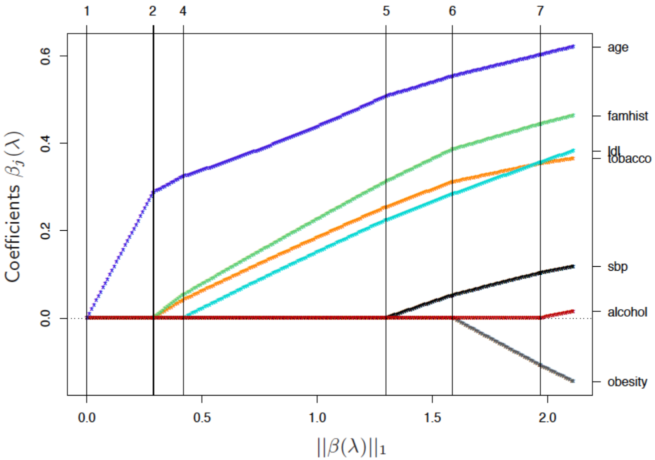


Figure: From ESL (online version). Note that the paths are not linear.

Linear Discriminant Analysis (LDA): Setup

- Suppose we have data from K groups (e.g., smokers vs. nonsmokers).
- We model predictors X as coming from a multivariate normal (MVN) distribution for each group:
$$X \sim MVN(\mu_k, \Sigma_k), \quad k = 1, \dots, K.$$
- If all groups share the same covariance matrix ($\Sigma_k = \Sigma$), then the boundaries between groups turn out to be **linear**.
- This is the key idea of **Linear Discriminant Analysis (LDA)**.

Linear Discriminant Function

- LDA builds a **discriminant function** for each group:

$$\delta_k(x) = x'\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k'\Sigma^{-1}\mu_k + \log \pi_k$$

  where
  - μ_k = mean vector for group k,
  - π_k = prior probability of being in group k,
  - Σ = common covariance matrix.
- The classification rule:

  Assign x to the group with the largest $\delta_k(x)$.

- In practice: this means each group is separated by a **linear boundary**.

Linear Discriminant Analysis Example

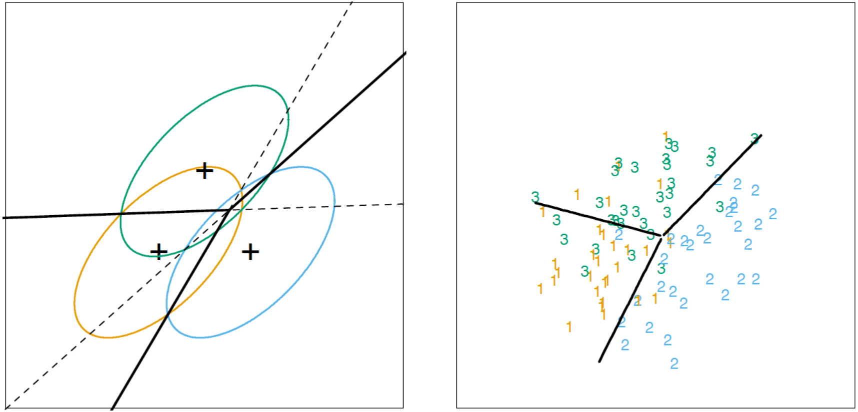


Figure: From ESL (online version).

Estimating the Parameters in LDA

- Estimate group priors: $\hat{\pi}_k = N_k/N$.
- Estimate group means: $\hat{\mu}_k =$ average of x_i in group k.
- Estimate pooled covariance: $\hat{\Sigma} =$ weighted average of within-group covariance.
- Note:
  - LDA does not rely heavily on the normality assumption for prediction.
  - The "cutpoint" (intercept) is more sensitive to distributional assumptions — often chosen or checked with cross-validation.

Quadratic Discriminant Analysis (QDA)

- If we allow each group to have its own covariance matrix (Σ_k), then the discriminant function becomes **quadratic**.
- The classification rule is still to assign x to the group with the largest $\delta_k(x)$, but the boundary curves can now be **non-linear**.
- Compared to LDA:
  - More flexible, captures complex separation between groups.
  - Requires estimating more parameters, so can overfit with small data.

LDA vs QDA: A Tradeoff

- **LDA**: simpler, linear boundaries, fewer parameters. Works well when groups are roughly separated by lines or planes.
- **QDA**: more flexible, quadratic boundaries, more parameters. Works well when groups need curved boundaries to be separated.
- Which to choose?
  - Try both and compare using cross-validation.
  - LDA is often more stable when sample size is small.